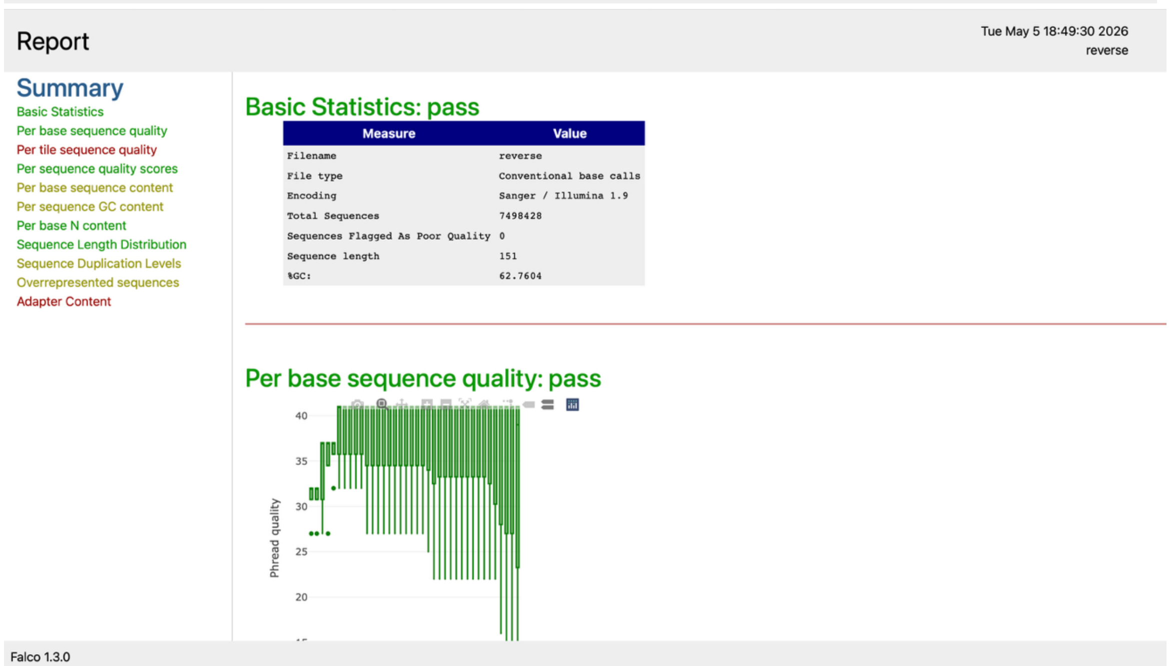
1. Falco reads the quality reports. It is an alternative to FastQC.
    a. I chose to analyze the SA52170 sample. Both forward and reverse reads were good quality, but there was room for some trimming. The adapter contamination didn't pass for both of them.

    b.



    c.

2. Fastp trims the reads and gets rid of anything that doesn't meet quality control.
    a. Analyzing the same sample (SA52170), the adapter sequences were removed. 96.7 % of the reads passed the filter. This ensures that the future alignment will be high quality and have no issues.

b.

SA52170.fastqsanger.gz

Summary

General

fastp version:	1.3.3 (https://github.com/OpenGene/fastp)
sequencing:	paired end (151 cycles + 151 cycles)
mean length before filtering:	151bp, 151bp
mean length after filtering:	144bp, 144bp
Insert size peak:	182

Before filtering

total reads:	14.996856 M
total bases:	2.264525 G
Q20 bases:	2.161762 G (95.462032%)
Q30 bases:	2.018136 G (89.119620%)
Q40 bases:	1.563408 G (69.039120%)
GC content:	62.693102%

After filtering

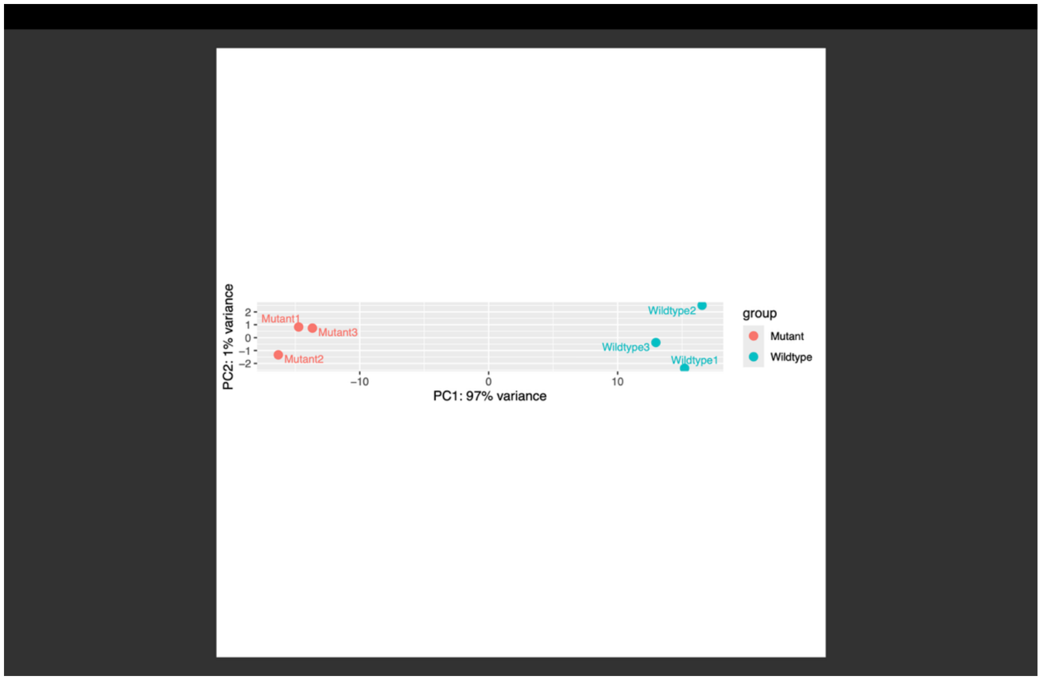
total reads:	14.516912 M
total bases:	2.096536 G
Q20 bases:	2.023518 G (96.517223%)
Q30 bases:	1.902400 G (90.740161%)
Q40 bases:	1.485741 G (70.866456%)
GC content:	63.346806%

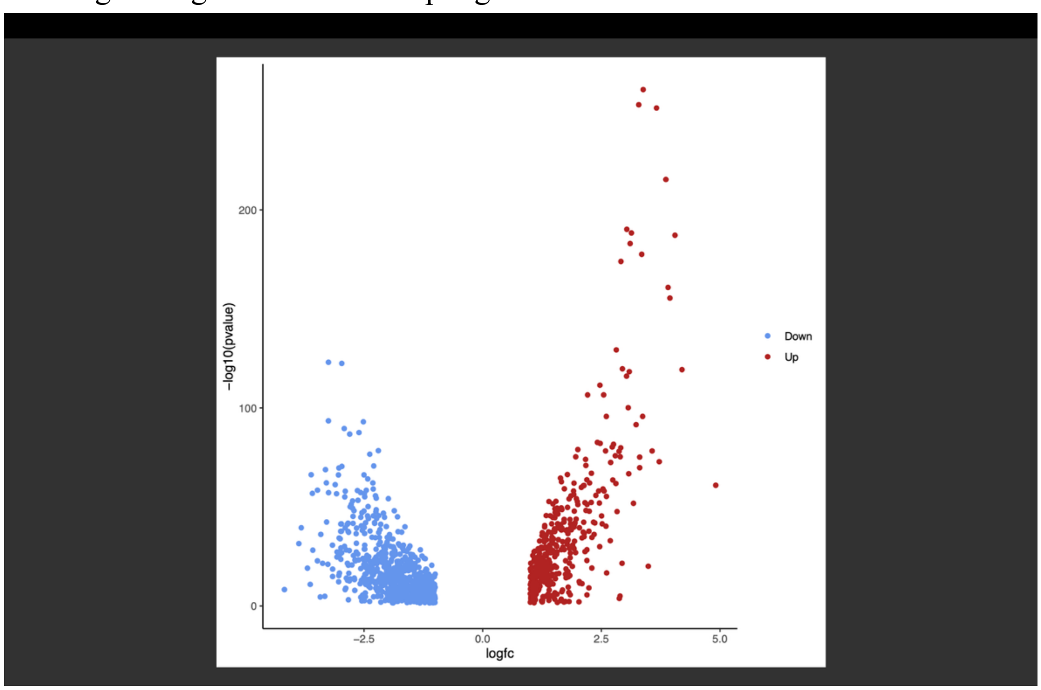
Filtering result

reads passed filters:	14.516912 M (96.799703%)
reads with low quality:	466.406000 K (3.110025%)
reads with too many N:	13.538000 K (0.090272%)
reads too short:	0 (0.000000%)
reads with adapter dimer:	0 (0.000000%)

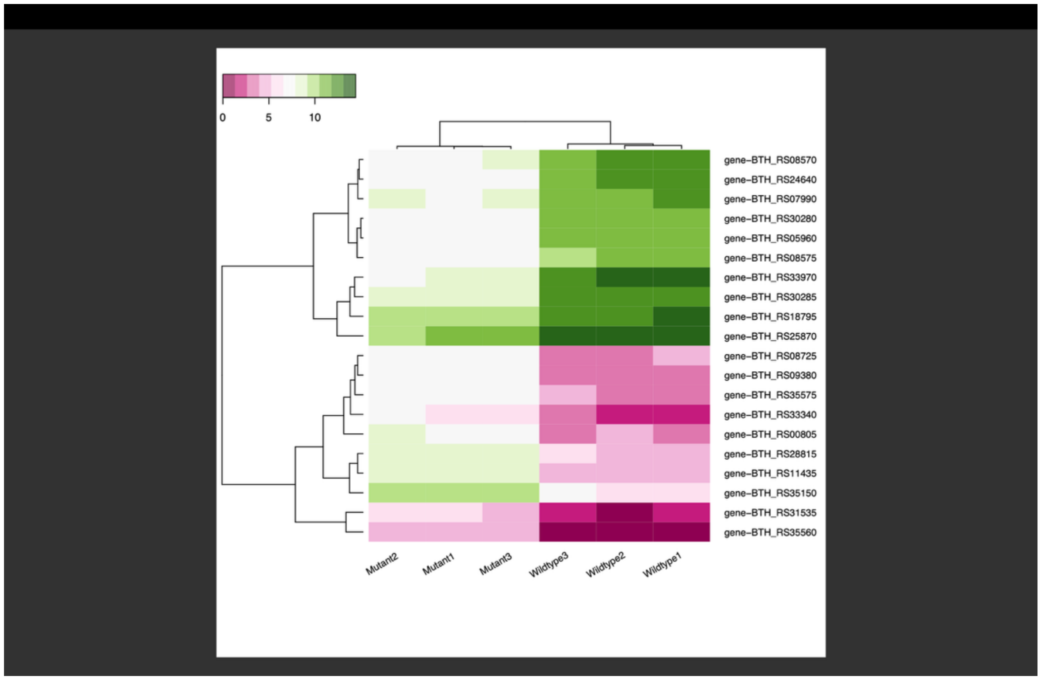
Adapters

3. HiSat2 acts as an alignment tool.
    a. It aligns my reference genome with the filtered dataset collection.
4. Stringtie assembles the short reads into transcripts.
    a. It takes my reference genome and HiSat2 results
5. featureCounts quantifies the read data by counting the sequence reads.
    a. It used by HiSat2 results and my stringtie merge.
6. DESeq2 determines differentially expressed features.
    a. It does this from my featureCounts table. The plot shows that PC1 accounts for 97% of the variance. This is a great result, as all the two types of genetic background are clumped with eachother. This means that there is a clear difference between wildtype and mutant and not just a random error.

b. 

7. Volcano Plot looks at changes within a large set of replicate data.
    a. This plot specifically looks at which genes changed the most and if they are statistically significant. The more left the gene is the more down regulated and the more right the gene is the more up regulated.
    b. 

8. Heatmap2 helps visualizes your data.
    a. The green shows a high level of expression, and the pink shows a low level of expression.

b. 

9. Up regulated gene: BTH_RS18795
    a. It has a positive log2 fold change of 3.49. This means that the gene is turned on a lot more than the wildtype. This gene is responsible in making Nitric Oxide Dioxygenase. Its job is to grab nitric oxide (which is toxic) and turn it to something harmless. This means that the mutant most likely has a better chance in surviving toxic environments than the wildtype.
10. Down regulated gene: BTH_RS31535
    a. It had a negative log2 fold change of -3.63. This means that the gene is turned off and slower than the wildtype. This gene is responsible in making Coproporphyrinogen III oxidase. Its job is to build molecules like heme. This might lead to a slower growth in the mutant compared to the wildtype.